

Module, UI Design & AI Integration

Project Title: TeamMatch

Introduction

TeamMatch is a web-based system designed to help instructors create balanced student teams for class projects. Instead of assigning teams manually, the system collects student information such as skills, schedules, experience, and preferences, then uses system rules and AI support to recommend better team groupings. The goal is to make team formation faster, fairer, and more effective while also improving team communication and project tracking.

Part 1: Module Design

1. User Management Module

Responsibility:

Handles user registration, login, authentication, profile creation, and account access. This module allows instructors and students to securely enter the system and use the features based on their role.

Main Functions:

- Register new users
- Login and logout
- Store user role information
- Manage profiles

Database Tables Used:

- User
- StudentProfile
- InstructorProfile

2. Team Formation Module

Responsibility:

Handles the core functionality of the system by creating teams based on student data, preferences, skills, and constraints. This module supports manual grouping, system-generated grouping, and AI-assisted recommendations.

Main Functions:

- Collect student responses
- Analyze skills and availability
- Generate team recommendations
- Create final teams

Database Tables Used:

- Team
 - TeamMember
 - StudentProfile
 - SurveyResponse
 - Skill
 - StudentSkill
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3. Project Management Module

Responsibility:

Manages the class project information connected to each team. It helps instructors assign projects and lets students view their assigned work.

Main Functions:

- Create project records
- Assign teams to projects
- Track project status
- Store deadlines and descriptions

Database Tables Used:

- Project
 - Team
 - Milestone
-

4. Progress Tracking Module

Responsibility:

Tracks how teams are performing during the project. This module records updates, participation, and milestone completion to help instructors monitor progress.

Main Functions:

- Submit progress updates
- Track completion of milestones
- View team activity
- Monitor project status

Database Tables Used:

- ProgressUpdate
 - Milestone
 - Team
 - Project
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5. Communication Module

Responsibility:

Supports notifications and communication between instructors and students. It helps users stay informed about team assignments, deadlines, and alerts.

Main Functions:

- Send notifications
- Display announcements
- Alert students about deadlines
- Inform users of team updates

Database Tables Used:

- Notification
 - Message
 - User
-

6. AI Recommendation Module

Responsibility:

Provides AI-generated team recommendations and team health insights. This module improves the quality of group formation by analyzing student data and predicting stronger team combinations.

Main Functions:

- Suggest balanced teams
- Flag risky team combinations
- Recommend improvements for weak teams
- Summarize team compatibility

Database Tables Used:

- AIRecommendation
 - StudentProfile
 - SurveyResponse
 - Team
 - TeamMember
 - StudentSkill
-

7. Admin and Reporting Module

Responsibility:

Allows instructors or administrators to review system results, reports, and team data. It supports decision-making and system monitoring.

Main Functions:

- View reports
- Export team results
- Monitor team balance

- Review AI suggestions

Database Tables Used:

- Report
 - Team
 - Project
 - User
 - AIRecommendation
-

Part 2: UI Design

1. Login Screen

Purpose:

Allows users to enter the system.

Components:

- Email field
 - Password field
 - Login button
 - Register button
 - Forgot password link
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2. Registration Screen

Purpose:

Allows new users to create an account.

Components:

- Name field
- Email field
- Password field
- Role selection dropdown

- Register button
 - Back to login link
-

3. Student Dashboard

Purpose:

Lets students manage their profile, skills, availability, and project information.

Components:

- Navigation sidebar
 - Profile summary card
 - Skills form
 - Availability form
 - Team information panel
 - Project updates section
 - Notifications panel
 - Submit button
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4. Instructor Dashboard

Purpose:

Lets instructors manage students, teams, and projects.

Components:

- Navigation sidebar
 - Student list table
 - Team generation button
 - AI recommendation panel
 - Project assignment table
 - Reports section
 - Notification panel
-

5. Team Formation Screen

Purpose:

Allows instructors to generate and review teams.

Components:

- Student selection table
 - Filters for skills and availability
 - Generate teams button
 - AI suggestion box
 - Team results table
 - Save teams button
-

6. Team Details Screen

Purpose:

Shows detailed information for a selected team.

Components:

- Team member list
 - Skill summary
 - Team compatibility score
 - Assigned project card
 - Progress tracker
 - Edit team button
-

7. Project Management Screen

Purpose:

Allows instructors to create and assign projects.

Components:

- Project creation form
- Team assignment dropdown
- Deadline field
- Milestone list
- Save button

- Project table
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8. Progress Tracking Screen

Purpose:

Lets teams submit updates and lets instructors review them.

Components:

- Update submission form
 - Milestone checklist
 - Progress history table
 - Status labels
 - Feedback section
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9. Reports Screen

Purpose:

Shows system reports and team performance summaries.

Components:

- Report filters
 - Team performance chart
 - Export button
 - AI insight summary
 - Completion statistics table
-

Part 3: Navigation Flow

Navigation Flow Description

The navigation flow of the system is designed to be simple and logical for both students and instructors.

Student Flow

Login → Student Dashboard → Complete Profile → Enter Skills and Availability → View Team Assignment → View Project → Submit Progress Update

Instructor Flow

Login → Instructor Dashboard → View Student Data → Open Team Formation Screen → Generate Teams → Review AI Suggestions → Save Teams → Assign Projects → Track Progress → View Reports

Diagram Format You Can Draw

You can create a simple diagram like this:

Login
↓
Dashboard
↓
Choose Role-Based Action
↓
Student: Profile / Team / Project / Progress
Instructor: Students / Team Formation / Projects / Reports

Part 4: AI Integration

AI Feature Name

AI Team Balancer

Problem It Solves

In many classrooms, team formation is done manually, which takes time and may create unbalanced groups. Some teams may end up with all strong students while others have

scheduling conflicts or missing skills. The AI Team Balancer helps solve this by recommending balanced teams based on student data.

Input

The AI feature uses:

- Student skills
- Availability
- Experience level
- Project preferences
- Survey responses
- Past participation data

Output

The AI generates:

- Recommended teams
- Team compatibility score
- Warnings for weak team combinations
- Suggestions for improving team balance

Where It Appears in the UI

The AI feature appears on the **Team Formation Screen** and also gives summaries on the **Instructor Dashboard**.

Which Module It Belongs To

This feature belongs to the **AI Recommendation Module**.

Part 5: System Mapping

UI to API to Database Mapping

1. Login Screen

UI Action: User enters email and password and clicks login

API: POST /login

Database Tables: User

2. Registration Screen

UI Action: User creates account

API: POST /register

Database Tables: User, StudentProfile, InstructorProfile

3. Student Profile Screen

UI Action: Student submits profile details

API: POST /student/profile

Database Tables: StudentProfile

4. Skills Submission

UI Action: Student adds technical and soft skills

API: POST /student/skills

Database Tables: Skill, StudentSkill

5. Availability Submission

UI Action: Student enters available times

API: POST /student/availability

Database Tables: SurveyResponse

6. Generate Teams

UI Action: Instructor clicks generate teams

API: POST /teams/generate

Database Tables: Team, TeamMember, StudentProfile, SurveyResponse, StudentSkill

7. View AI Suggestions

UI Action: Instructor requests AI team recommendation

API: GET /ai/team-recommendations

Database Tables: AIRecommendation, StudentProfile, SurveyResponse, StudentSkill

8. Save Final Teams

UI Action: Instructor confirms generated teams

API: POST /teams/save

Database Tables: Team, TeamMember

9. Assign Project

UI Action: Instructor assigns a project to a team

API: POST /projects/assign

Database Tables: Project, Team

10. Submit Progress Update

UI Action: Team submits update

API: POST /progress/update

Database Tables: ProgressUpdate, Milestone, Team

11. View Reports

UI Action: Instructor opens reports page

API: GET /reports

Database Tables: Report, Team, Project, AIRecommendation

Part 6: Existing System Analysis

Existing Application

CATME

Key Features of the Current System

CATME is a teamwork management system used in education. Its Team-Maker feature helps instructors form teams using selected criteria, and its peer evaluation tools allow students to evaluate one another using teamwork dimensions. CATME also allows instructors to view team results and rerun team formation using different criteria.

Limitations or Gaps in the Current System

Although CATME is strong for team formation and peer evaluation, the official materials emphasize instructor-selected team criteria and survey-based team creation rather than a more modern dashboard experience with project tracking, AI-generated team health insights, and an integrated workflow from team creation through project progress monitoring. CATME materials also note that, for Team-Maker results released to students, the information is limited mainly to teammate contact and schedule availability rather than a broader collaboration workspace.

Comparison with Proposed System

Our proposed system, TeamMatch, is similar to CATME because both systems focus on building better student teams. However, TeamMatch improves the experience by combining team formation, project management, progress tracking, and AI support in one system.

CATME mainly focuses on forming teams and evaluating teamwork behavior. TeamMatch goes further by helping instructors manage the full project lifecycle after the team is created. It also gives students a clearer dashboard where they can see their team, assigned project, deadlines, and progress updates in one place.

What Makes Our System Better

TeamMatch is better because it introduces a more complete and modern workflow. Instead of only forming teams, it also helps instructors assign projects, monitor milestones, and receive AI-based suggestions about team balance and possible risks. This makes the system more useful across the entire semester, not just at the team creation stage.

New Features or Improvements Introduced

The new features in TeamMatch include:

- AI-generated team balancing suggestions
- Team compatibility scoring
- Progress tracking after team creation
- Project assignment tools
- Instructor dashboard with reports and alerts
- Student dashboard with project and team visibility

Why Our Solution Is More Efficient, Modern, and User-Friendly

TeamMatch is more efficient because it reduces manual work for instructors by automating team recommendations and organizing project tracking in one place. It is more modern because it includes AI integration and dashboard-based system design. It is more user-friendly because both students and instructors have role-based screens that match what they actually need to do in the system.

Conclusion

In conclusion, TeamMatch is designed as a structured and intelligent team formation system for education. The modules clearly separate responsibilities such as user management, team formation, project tracking, communication, and AI support. The UI design provides simple navigation for both students and instructors, while the system mapping shows how the interface connects to APIs and database tables. By comparing the project to an existing system like CATME, it becomes clear that TeamMatch offers a more complete, modern, and practical solution for managing student teams and projects.